

xtreg — Fixed-, between-, and random-effects and population-averaged linear models

Syntax	Menu	Description
Options for RE model	Options for BE model	Options for FE model
Options for MLE model	Options for PA model	Remarks and examples
Stored results	Methods and formulas	Acknowledgments
References	Also see	

Syntax

GLS random-effects (RE) model

```
xtreg depvar [indepvars] [if] [in] [, re RE_options]
```

Between-effects (BE) model

```
xtreg depvar [indepvars] [if] [in] , be [BE_options]
```

Fixed-effects (FE) model

```
xtreg depvar [indepvars] [if] [in] [weight] , fe [FE_options]
```

ML random-effects (MLE) model

```
xtreg depvar [indepvars] [if] [in] [weight] , mle [MLE_options]
```

Population-averaged (PA) model

```
xtreg depvar [indepvars] [if] [in] [weight] , pa [PA_options]
```

RE_options	Description
Model	
re	use random-effects estimator; the default
sa	use Swamy–Arora estimator of the variance components
SE/Robust	
vce(vcetype)	vcetype may be conventional, <u>robust</u> , <u>cluster</u> <i>clustvar</i> , <u>bootstrap</u> , or <u>jackknife</u>
Reporting	
level(#)	set confidence level; default is level(95)
theta	report θ
display_options	control column formats, row spacing, line width, display of omitted variables and base and empty cells, and factor-variable labeling
coeflegend	display legend instead of statistics

<i>BE_options</i>	Description
Model	
be	use between-effects estimator
wls	use weighted least squares
SE	
vce (<i>vcetype</i>)	<i>vcetype</i> may be conventional, <u>bootstrap</u> , or <u>jackknife</u>
Reporting	
level (#)	set confidence level; default is <code>level(95)</code>
<i>display_options</i>	control column formats, row spacing, line width, display of omitted variables and base and empty cells, and factor-variable labeling
coeflegend	display legend instead of statistics
<i>FE_options</i>	Description
Model	
fe	use fixed-effects estimator
SE/Robust	
vce (<i>vcetype</i>)	<i>vcetype</i> may be conventional, <u>robust</u> , <u>cluster</u> <i>clustvar</i> , <u>bootstrap</u> , or <u>jackknife</u>
Reporting	
level (#)	set confidence level; default is <code>level(95)</code>
<i>display_options</i>	control column formats, row spacing, line width, display of omitted variables and base and empty cells, and factor-variable labeling
coeflegend	display legend instead of statistics
<i>MLE_options</i>	Description
Model	
noconstant	suppress constant term
mle	use ML random-effects estimator
SE	
vce (<i>vcetype</i>)	<i>vcetype</i> may be <code>oim</code> , <u>bootstrap</u> , or <u>jackknife</u>
Reporting	
level (#)	set confidence level; default is <code>level(95)</code>
<i>display_options</i>	control column formats, row spacing, line width, display of omitted variables and base and empty cells, and factor-variable labeling
Maximization	
<i>maximize_options</i>	control the maximization process; seldom used
coeflegend	display legend instead of statistics

<i>PA_options</i>	Description
Model	
<u>noconstant</u>	suppress constant term
<u>pa</u>	use population-averaged estimator
<u>offset</u> (<i>varname</i>)	include <i>varname</i> in model with coefficient constrained to 1
Correlation	
<u>corr</u> (<i>correlation</i>)	within-panel correlation structure
<u>force</u>	estimate even if observations unequally spaced in time
SE/Robust	
<u>vce</u> (<i>vcetype</i>)	<i>vcetype</i> may be <u>conventional</u> , <u>robust</u> , <u>bootstrap</u> , or <u>jackknife</u>
<u>nmp</u>	use divisor $N - P$ instead of the default N
<u>rgf</u>	multiply the robust variance estimate by $(N - 1)/(N - P)$
<u>scale</u> (<i>parm</i>)	overrides the default scale parameter; <i>parm</i> may be <i>x2</i> , <i>dev</i> , <i>phi</i> , or <i>#</i>
Reporting	
<u>level</u> (<i>#</i>)	set confidence level; default is <code>level(95)</code>
<u>display_options</u>	control column formats, row spacing, line width, display of omitted variables and base and empty cells, and factor-variable labeling
Optimization	
<u>optimize_options</u>	control the optimization process; seldom used
<u>coeflegend</u>	display legend instead of statistics
<i>correlation</i>	Description
<u>exchangeable</u>	exchangeable
<u>independent</u>	independent
<u>unstructured</u>	unstructured
<u>fixed</u> <i>matname</i>	user-specified
<u>ar</u> <i>#</i>	autoregressive of order <i>#</i>
<u>stationary</u> <i>#</i>	stationary of order <i>#</i>
<u>nonstationary</u> <i>#</i>	nonstationary of order <i>#</i>

A panel variable must be specified. For `xtreg`, `pa`, correlation structures other than `exchangeable` and `independent` require that a time variable also be specified. Use `xtset`; see [XT] [xtset](#).

indepvars may contain factor variables; see [U] [11.4.3 Factor variables](#).

depvar and *indepvars* may contain time-series operators; see [U] [11.4.4 Time-series varlists](#).

`by`, `mi estimate`, and `statsby` are allowed; see [U] [11.1.10 Prefix commands](#). `fp` is allowed for the between-effects, fixed-effects, and maximum-likelihood random-effects models.

`vce(bootstrap)` and `vce(jackknife)` are not allowed with the `mi estimate` prefix; see [MI] [mi estimate](#).

`aweight`s, `fweight`s, and `pweight`s are allowed for the fixed-effects model. `iweight`s, `fweight`s, and `pweight`s are allowed for the population-averaged model. `iweight`s are allowed for the maximum-likelihood random-effects (MLE) model. See [U] [11.1.6 weight](#). Weights must be constant within panel.

`coeflegend` does not appear in the dialog box.

See [U] [20 Estimation and postestimation commands](#) for more capabilities of estimation commands.

Menu

Statistics > Longitudinal/panel data > Linear models > Linear regression (FE, RE, PA, BE)

Description

xtreg fits regression models to panel data. In particular, **xtreg** with the **be** option fits random-effects models by using the between regression estimator; with the **fe** option, it fits fixed-effects models (by using the within regression estimator); and with the **re** option, it fits random-effects models by using the GLS estimator (producing a matrix-weighted average of the between and within results). See [XT] **xtdata** for a faster way to fit fixed- and random-effects models.

Options for RE model

Model

re, the default, requests the GLS random-effects estimator.

sa specifies that the small-sample Swamy–Arora estimator individual-level variance component be used instead of the default consistent estimator. See **xtreg**, **re** in *Methods and formulas* for details.

SE/Robust

vce(vctype) specifies the type of standard error reported, which includes types that are derived from asymptotic theory (**conventional**), that are robust to some kinds of misspecification (**robust**), that allow for intragroup correlation (**cluster clustvar**), and that use bootstrap or jackknife methods (**bootstrap**, **jackknife**); see [XT] **vce_options**.

vce(conventional), the default, uses the conventionally derived variance estimator for generalized least-squares regression.

Specifying **vce(robust)** is equivalent to specifying **vce(cluster panelvar)**; see **xtreg**, **re** in *Methods and formulas*.

Reporting

level(#); see [R] **estimation options**.

theta specifies that the output include the estimated value of θ used in combining the between and fixed estimators. For balanced data, this is a constant, and for unbalanced data, a summary of the values is presented in the header of the output.

display_options: **noomitted**, **vsquish**, **noemptycells**, **baselevels**, **allbaselevels**, **nofvlabel**, **fvwrap(#)**, **fvwrapon(style)**, **cformat(%fmt)**, **pformat(%fmt)**, **sformat(%fmt)**, and **nolstretch**; see [R] **estimation options**.

The following option is available with **xtreg** but is not shown in the dialog box:

coeflegend; see [R] **estimation options**.

Options for BE model

Model

`be` requests the between regression estimator.

`wls` specifies that, for unbalanced data, weighted least squares be used rather than the default OLS.

Both methods produce consistent estimates. The true variance of the between-effects residual is $\sigma_\nu^2 + T_i\sigma_\epsilon^2$ (see [xtreg](#), [be](#) in *Methods and formulas* below). WLS produces a “stabilized” variance of $\sigma_\nu^2/T_i + \sigma_\epsilon^2$, which is also not constant. Thus the choice between OLS and WLS amounts to which is more stable.

Comment: `xtreg`, `be` is rarely used anyway, but between estimates are an ingredient in the random-effects estimate. Our implementation of `xtreg`, `re` uses the OLS estimates for this ingredient, based on our judgment that σ_ν^2 is large relative to σ_ϵ^2 in most models. Formally, only a consistent estimate of the between estimates is required.

SE

`vce(vcetype)` specifies the type of standard error reported, which includes types that are derived from asymptotic theory (`conventional`) and that use bootstrap or jackknife methods (`bootstrap`, `jackknife`); see [\[XT\] vce_options](#).

`vce(conventional)`, the default, uses the conventionally derived variance estimator for generalized least-squares regression.

Reporting

`level(#)`; see [\[R\] estimation options](#).

`display_options`: `noomitted`, `vsquish`, `noemptycells`, `baselevels`, `allbaselevels`, `nofvlabel`, `fvwrap(#)`, `fvwrapon(style)`, `cformat(%fint)`, `pformat(%fint)`, `sformat(%fint)`, and `no!stretch`; see [\[R\] estimation options](#).

The following option is available with `xtreg` but is not shown in the dialog box:

`coeflegend`; see [\[R\] estimation options](#).

Options for FE model

Model

`fe` requests the fixed-effects (within) regression estimator.

SE/Robust

`vce(vcetype)` specifies the type of standard error reported, which includes types that are derived from asymptotic theory (`conventional`), that are robust to some kinds of misspecification (`robust`), that allow for intragroup correlation (`cluster clustvar`), and that use bootstrap or jackknife methods (`bootstrap`, `jackknife`); see [\[XT\] vce_options](#).

`vce(conventional)`, the default, uses the conventionally derived variance estimator for generalized least-squares regression.

Specifying `vce(robust)` is equivalent to specifying `vce(cluster panelvar)`; see [xtreg](#), [fe](#) in *Methods and formulas*.

Reporting

`level(#)`; see [R] [estimation options](#).

display_options: `noomitted`, `vsquish`, `noemptycells`, `baselevels`, `allbaselevels`, `nofvlabel`, `fvwrap(#)`, `fvwrapon(style)`, `cformat(%fmt)`, `pformat(%fmt)`, `sformat(%fmt)`, and `nolstretch`; see [R] [estimation options](#).

The following option is available with `xtreg` but is not shown in the dialog box:

`coeflegend`; see [R] [estimation options](#).

Options for MLE model

Model

`noconstant`; see [R] [estimation options](#).

`mle` requests the maximum-likelihood random-effects estimator.

SE

`vce(vcetype)` specifies the type of standard error reported, which includes types that are derived from asymptotic theory (`oim`) and that use bootstrap or jackknife methods (`bootstrap`, `jackknife`); see [XT] [vce_options](#).

Reporting

`level(#)`; see [R] [estimation options](#).

display_options: `noomitted`, `vsquish`, `noemptycells`, `baselevels`, `allbaselevels`, `nofvlabel`, `fvwrap(#)`, `fvwrapon(style)`, `cformat(%fmt)`, `pformat(%fmt)`, `sformat(%fmt)`, and `nolstretch`; see [R] [estimation options](#).

Maximization

maximize_options: `iterate(#)`, `[no]log`, `trace`, `tolerance(#)`, `ltolerance(#)`, and `from(init_specs)`; see [R] [maximize](#). These options are seldom used.

The following option is available with `xtreg` but is not shown in the dialog box:

`coeflegend`; see [R] [estimation options](#).

Options for PA model

Model

`noconstant`; see [R] [estimation options](#).

`pa` requests the population-averaged estimator. For linear regression, this is the same as a random-effects estimator (both interpretations hold).

`xtreg, pa` is equivalent to `xtgee, family(gaussian) link(id) corr(exchangeable)`, which are the defaults for the `xtgee` command. `xtreg, pa` allows all the relevant `xtgee` options such as `vce(robust)`. Whether you use `xtreg, pa` or `xtgee` makes no difference. See [XT] [xtgee](#).

`offset(varname)`; see [R] [estimation options](#).

Correlation

`corr(correlation)` specifies the within-panel correlation structure; the default corresponds to the equal-correlation model, `corr(exchangeable)`.

When you specify a correlation structure that requires a lag, you indicate the lag after the structure's name with or without a blank; for example, `corr(ar 1)` or `corr(ar1)`.

If you specify the fixed correlation structure, you specify the name of the matrix containing the assumed correlations following the word `fixed`, for example, `corr(fixed myr)`.

`force` specifies that estimation be forced even though the time variable is not equally spaced.

This is relevant only for correlation structures that require knowledge of the time variable. These correlation structures require that observations be equally spaced so that calculations based on lags correspond to a constant time change. If you specify a time variable indicating that observations are not equally spaced, the (time dependent) model will not be fit. If you also specify `force`, the model will be fit, and it will be assumed that the lags based on the data ordered by the time variable are appropriate.

SE/Robust

`vce(vctype)` specifies the type of standard error reported, which includes types that are derived from asymptotic theory (`conventional`), that are robust to some kinds of misspecification (`robust`), and that use bootstrap or jackknife methods (`bootstrap`, `jackknife`); see [XT] [vce_options](#).

`vce(conventional)`, the default, uses the conventionally derived variance estimator for generalized least-squares regression.

`nmp`; see [XT] [vce_options](#).

`rgf` specifies that the robust variance estimate is multiplied by $(N - 1)/(N - P)$, where N is the total number of observations and P is the number of coefficients estimated. This option can be used with `family(gaussian)` only when `vce(robust)` is either specified or implied by the use of `pweights`. Using this option implies that the robust variance estimate is not invariant to the scale of any weights used.

`scale(x2 | dev | phi | #)`; see [XT] [vce_options](#).

Reporting

`level(#)`; see [R] [estimation_options](#).

`display_options`: `noomitted`, `vsquish`, `noemptycells`, `baselevels`, `allbaselevels`, `novlabel`, `fvwrap(#)`, `fvwrapon(style)`, `cformat(%fmt)`, `pformat(%fmt)`, `sformat(%fmt)`, and `nolstretch`; see [R] [estimation_options](#).

Optimization

`optimize_options` control the iterative optimization process. These options are seldom used.

`iterate(#)` specifies the maximum number of iterations. When the number of iterations equals `#`, the optimization stops and presents the current results, even if convergence has not been reached. The default is `iterate(100)`.

`tolerance(#)` specifies the tolerance for the coefficient vector. When the relative change in the coefficient vector from one iteration to the next is less than or equal to `#`, the optimization process is stopped. `tolerance(1e-6)` is the default.

`nolog` suppresses display of the iteration log.

`trace` specifies that the current estimates be printed at each iteration.

The following option is available with **xtreg** but is not shown in the dialog box: `coeflegend`; see [R] [estimation options](#).

Remarks and examples

[stata.com](http://www.stata.com)

If you have not read [XT] **xt**, please do so.

See Baltagi (2013, chap. 2) and Wooldridge (2013, chap. 14) for good overviews of fixed-effects and random-effects models. Allison (2009) provides perspective on the use of fixed- versus random-effects estimators and provides many examples using Stata.

Consider fitting models of the form

$$y_{it} = \alpha + \mathbf{x}_{it}\beta + \nu_i + \epsilon_{it} \quad (1)$$

In this model, $\nu_i + \epsilon_{it}$ is the error term that we have little interest in; we want estimates of β . ν_i is the unit-specific error term; it differs between units, but for any particular unit, its value is constant. In the pulmonary data of [XT] **xt**, a person who exercises less would presumably have a lower forced expiratory volume (FEV) year after year and so would have a negative ν_i .

ϵ_{it} is the “usual” error term with the usual properties (mean 0, uncorrelated with itself, uncorrelated with $\mathbf{x}$, uncorrelated with ν , and homoskedastic), although in a more thorough development, we could decompose $\epsilon_{it} = v_t + \omega_{it}$, assume that ω_{it} is a conventional error term, and better describe v_t .

Before making the assumptions necessary for estimation, let’s perform some useful algebra on (1). Whatever the properties of ν_i and ϵ_{it} , if (1) is true, it must also be true that

$$\bar{y}_i = \alpha + \bar{\mathbf{x}}_i\beta + \nu_i + \bar{\epsilon}_i \quad (2)$$

where $\bar{y}_i = \sum_t y_{it}/T_i$, $\bar{\mathbf{x}}_i = \sum_t \mathbf{x}_{it}/T_i$, and $\bar{\epsilon}_i = \sum_t \epsilon_{it}/T_i$. Subtracting (2) from (1), it must be equally true that

$$(y_{it} - \bar{y}_i) = (\mathbf{x}_{it} - \bar{\mathbf{x}}_i)\beta + (\epsilon_{it} - \bar{\epsilon}_i) \quad (3)$$

These three equations provide the basis for estimating β . In particular, **xtreg**, **fe** provides what is known as the fixed-effects estimator—also known as the within estimator—and amounts to using OLS to perform the estimation of (3). **xtreg**, **be** provides what is known as the between estimator and amounts to using OLS to perform the estimation of (2). **xtreg**, **re** provides the random-effects estimator and is a (matrix) weighted average of the estimates produced by the between and within estimators. In particular, the random-effects estimator turns out to be equivalent to estimation of

$$(y_{it} - \theta\bar{y}_i) = (1 - \theta)\alpha + (\mathbf{x}_{it} - \theta\bar{\mathbf{x}}_i)\beta + \{(1 - \theta)\nu_i + (\epsilon_{it} - \theta\bar{\epsilon}_i)\} \quad (4)$$

where θ is a function of σ_ν^2 and σ_ϵ^2 . If $\sigma_\nu^2 = 0$, meaning that ν_i is always 0, $\theta = 0$ and (1) can be estimated by OLS directly. Alternatively, if $\sigma_\epsilon^2 = 0$, meaning that ϵ_{it} is 0, $\theta = 1$ and the within estimator returns all the information available (which will, in fact, be a regression with an R^2 of 1).

For more reasonable cases, few assumptions are required to justify the fixed-effects estimator of (3). The estimates are, however, conditional on the sample in that the ν_i are not assumed to have a distribution but are instead treated as fixed and estimable. This statistical fine point can lead to difficulty when making out-of-sample predictions, but that aside, the fixed-effects estimator has much to recommend it.

More is required to justify the between estimator of (2), but the conditioning on the sample is not assumed because $\nu_i + \bar{\epsilon}_i$ is treated as an error term. Newly required is that we assume that ν_i and $\bar{\mathbf{x}}_i$ are uncorrelated. This follows from the assumptions of the OLS estimator but is also transparent: were ν_i and $\bar{\mathbf{x}}_i$ correlated, the estimator could not determine how much of the change in $\bar{y}_i$, associated with an increase in $\bar{\mathbf{x}}_i$, to assign to β versus how much to attribute to the unknown correlation. (This, of course, suggests the use of an instrumental-variable estimator, $\bar{\mathbf{z}}_i$, which is correlated with $\bar{\mathbf{x}}_i$ but uncorrelated with ν_i , though that approach is not implemented here.)

The random-effects estimator of (4) requires the same no-correlation assumption. In comparison with the between estimator, the random-effects estimator produces more efficient results, albeit ones with unknown small-sample properties. The between estimator is less efficient because it discards the over-time information in the data in favor of simple means; the random-effects estimator uses both the within and the between information.

All this would seem to leave the between estimator of (2) with no role (except for a minor, technical part it plays in helping to estimate σ_ν^2 and σ_ϵ^2 , which are used in the calculation of θ , on which the random-effects estimates depend). Let's, however, consider a variation on (1):

$$y_{it} = \alpha + \bar{\mathbf{x}}_i\beta_1 + (\mathbf{x}_{it} - \bar{\mathbf{x}}_i)\beta_2 + \nu_i + \epsilon_{it} \quad (1')$$

In this model, we postulate that changes in the average value of $\mathbf{x}$ for an individual have a different effect from temporary departures from the average. In an economic situation, y might be purchases of some item and $\mathbf{x}$ income; a change in average income should have more effect than a transitory change. In a clinical situation, y might be a physical response and $\mathbf{x}$ the level of a chemical in the brain; the model allows a different response to permanent rather than transitory changes.

The variations of (2) and (3) corresponding to (1') are

$$\bar{y}_i = \alpha + \bar{\mathbf{x}}_i\beta_1 + \nu_i + \bar{\epsilon}_i \quad (2')$$

$$(y_{it} - \bar{y}_i) = (\mathbf{x}_{it} - \bar{\mathbf{x}}_i)\beta_2 + (\epsilon_{it} - \bar{\epsilon}_i) \quad (3')$$

That is, the between estimator estimates β_1 and the within β_2 , and neither estimates the other. Thus even when estimating equations like (1), it is worth comparing the within and between estimators. Differences in results can suggest models like (1'), or at the least some other specification error.

Finally, it is worth understanding the role of the between and within estimators with regressors that are constant over time or constant over units. Consider the model

$$y_{it} = \alpha + \mathbf{x}_{it}\beta_1 + \mathbf{s}_i\beta_2 + \mathbf{z}_t\beta_3 + \nu_i + \epsilon_{it} \quad (1'')$$

This model is the same as (1), except that we explicitly identify the variables that vary over both time and i ($\mathbf{x}_{it}$, such as output or FEV); variables that are constant over time ($\mathbf{s}_i$, such as race or sex); and variables that vary solely over time ($\mathbf{z}_t$, such as the consumer price index or age in a cohort study). The corresponding between and within equations are

$$\bar{y}_i = \alpha + \bar{\mathbf{x}}_i\beta_1 + \mathbf{s}_i\beta_2 + \bar{\mathbf{z}}\beta_3 + \nu_i + \bar{\epsilon}_i \quad (2'')$$

$$(y_{it} - \bar{y}_i) = (\mathbf{x}_{it} - \bar{\mathbf{x}}_i)\beta_1 + (\mathbf{z}_t - \bar{\mathbf{z}})\beta_3 + (\epsilon_{it} - \bar{\epsilon}_i) \quad (3'')$$

In the between estimator of (2''), no estimate of β_3 is possible because $\bar{\mathbf{z}}$ is a constant across the i observations; the regression-estimated intercept will be an estimate of $\alpha + \bar{\mathbf{z}}\beta_3$. On the other hand, it can provide estimates of β_1 and β_2 . It can estimate effects of factors that are constant over time, such as race and sex, but to do so it must assume that ν_i is uncorrelated with those factors.

The within estimator of (3''), like the between estimator, provides an estimate of β_1 but provides no estimate of β_2 for time-invariant factors. Instead, it provides an estimate of β_3 , the effects of the time-varying factors. The within estimator can also provide estimates u_i for ν_i . More correctly, the estimator u_i is an estimator of $\nu_i + \mathbf{s}_i\beta_2$. Thus u_i is an estimator of ν_i only if there are no time-invariant variables in the model. If there are time-invariant variables, u_i is an estimate of ν_i plus the effects of the time-invariant variables.

Remarks are presented under the following headings:

Assessing goodness of fit
xtreg and associated commands

Assessing goodness of fit

R^2 is a popular measure of goodness of fit in ordinary regression. In our case, given $\hat{\alpha}$ and $\hat{\beta}$ estimates of α and β , we can assess the goodness of fit with respect to (1), (2), or (3). The prediction equations are, respectively,

$$\hat{y}_{it} = \hat{\alpha} + \mathbf{x}_{it}\hat{\beta} \quad (1''')$$

$$\hat{\bar{y}}_i = \hat{\alpha} + \bar{\mathbf{x}}_i\hat{\beta} \quad (2''')$$

$$\hat{\tilde{y}}_{it} = (\hat{y}_{it} - \hat{\bar{y}}_i) = (\mathbf{x}_{it} - \bar{\mathbf{x}}_i)\hat{\beta} \quad (3''')$$

xtreg reports “ R -squares” corresponding to these three equations. R -squares is in quotes because the R -squares reported do not have all the properties of the OLS R^2 .

The ordinary properties of R^2 include being equal to the squared correlation between $\hat{y}$ and y and being equal to the fraction of the variation in y explained by $\hat{y}$ —formally defined as $\text{Var}(\hat{y})/\text{Var}(y)$. The identity of the definitions is from a special property of the OLS estimates; in general, given a prediction $\hat{y}$ for y , the squared correlation is not equal to the ratio of the variances, and the ratio of the variances is not required to be less than 1.

xtreg reports R^2 values calculated as correlations squared, calling them R^2 overall, corresponding to (1'''); R^2 between, corresponding to (2'''); and R^2 within, corresponding to (3'''). In fact, you can think of each of these three numbers as having all the properties of ordinary R^2 s, if you bear in mind that the prediction being judged is not $\hat{y}_{it}$, $\hat{\bar{y}}_i$, and $\hat{\tilde{y}}_{it}$, but $\gamma_1\hat{y}_{it}$ from the regression $y_{it} = \gamma_1\hat{y}_{it}$; $\gamma_2\hat{\bar{y}}_i$ from the regression $\bar{y}_i = \gamma_2\hat{\bar{y}}_i$; and $\gamma_3\hat{\tilde{y}}_{it}$ from $\tilde{y}_{it} = \gamma_3\hat{\tilde{y}}_{it}$.

In particular, **xtreg, be** obtains its estimates by performing OLS on (2), and therefore its reported R^2 between is an ordinary R^2 . The other two reported R^2 s are merely correlations squared, or, if you prefer, R^2 s from the second-round regressions $y_{it} = \gamma_{11}\hat{y}_{it}$ and $\tilde{y}_{it} = \gamma_{13}\hat{\tilde{y}}_{it}$.

xtreg, fe obtains its estimates by performing OLS on (3), so its reported R^2 within is an ordinary R^2 . As with **be**, the other R^2 s are correlations squared, or, if you prefer, R^2 s from the second-round regressions $\bar{y}_i = \gamma_{22}\hat{\bar{y}}_i$ and, as with **be**, $\tilde{y}_{it} = \gamma_{23}\hat{\tilde{y}}_{it}$.

xtreg, re obtains its estimates by performing OLS on (4); none of the R^2 s corresponding to (1'''), (2'''), or (3''') correspond directly to this estimator (the “relevant” R^2 is the one corresponding to (4)). All three reported R^2 s are correlations squared, or, if you prefer, from second-round regressions.

xtreg and associated commands

► Example 1: Between-effects model

Using `nlswork.dta` described in [XT] `xt`, we will model `ln_wage` in terms of completed years of schooling (`grade`), current age and age squared, current years worked (`experience`) and `experience` squared, current years of tenure on the current job and `tenure` squared, whether black (`race = 2`), whether residing in an area not designated a standard metropolitan statistical area (SMSA), and whether residing in the South.

```
. use http://www.stata-press.com/data/r13/nlswork
(National Longitudinal Survey. Young Women 14-26 years of age in 1968)
```

To obtain the between-effects estimates, we use `xtreg, be`. `nlswork.dta` has previously been `xtset idcode year` because that is what is true of the data, but for running `xtreg`, it would have been sufficient to have `xtset idcode` by itself.

```
. xtreg ln_w grade age c.age#c.age ttl_exp c.ttl_exp#c.ttl_exp tenure
> c.tenure#c.tenure 2.race not_smsa south, be

Between regression (regression on group means)   Number of obs       =      28091
Group variable: idcode                         Number of groups     =      4697

R-sq:  within = 0.1591                          Obs per group: min =         1
        between = 0.4900                          avg =         6.0
        overall = 0.3695                          max =        15

                                                F(10,4686)          =      450.23
                                                Prob > F             =      0.0000

sd(u_i + avg(e_i.))= .3036114
```

ln_wage	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
grade	.0607602	.0020006	30.37	0.000	.0568382	.0646822
age	.0323158	.0087251	3.70	0.000	.0152105	.0494211
c.age#c.age	-.0005997	.0001429	-4.20	0.000	-.0008799	-.0003194
ttl_exp	.0138853	.0056749	2.45	0.014	.0027598	.0250108
c.ttl_exp#						
c.ttl_exp	.0007342	.0003267	2.25	0.025	.0000936	.0013747
tenure	.0698419	.0060729	11.50	0.000	.0579361	.0817476
c.tenure#						
c.tenure	-.0028756	.0004098	-7.02	0.000	-.0036789	-.0020722
race						
black	-.0564167	.0105131	-5.37	0.000	-.0770272	-.0358061
not_smsa	-.1860406	.0112495	-16.54	0.000	-.2080949	-.1639862
south	-.0993378	.010136	-9.80	0.000	-.1192091	-.0794665
_cons	.3339113	.1210434	2.76	0.006	.0966093	.5712133

The between-effects regression is estimated on person-averages, so the “ $n = 4697$ ” result is relevant. `xtreg, be` reports the “number of observations” and group-size information: `describe` in [XT] `xt` showed that we have 28,534 “observations”—person-years, really—of data. If we take the subsample that has no missing values in `ln_wage`, `grade`, ..., `south` leaves us with 28,091 observations on person-years, reflecting 4,697 persons, each observed for an average of 6.0 years.

For goodness of fit, the R^2 between is directly relevant; our R^2 is 0.4900. If, however, we use these estimates to predict the within model, we have an R^2 of 0.1591. If we use these estimates to fit the overall data, our R^2 is 0.3695.

The F statistic tests that the coefficients on the regressors `grade`, `age`, `...`, `south` are all jointly zero. Our model is significant.

The root mean squared error of the fitted regression, which is an estimate of the standard deviation of $\nu_i + \bar{\epsilon}_i$, is 0.3036.

For our coefficients, each year of schooling increases hourly wages by 6.1%; age increases wages up to age 26.9 and thereafter decreases them (because the quadratic $ax^2 + bx + c$ turns over at $x = -b/2a$, which for our `age` and `c.age#c.age` coefficients is $0.0323158/(2 \times 0.0005997) \approx 26.9$); total experience increases wages at an increasing rate (which is surprising and bothersome); tenure on the current job increases wages up to a tenure of 12.1 years and thereafter decreases them; wages of blacks are, these things held constant, (approximately) 5.6% below that of nonblacks (approximately because `2.race` is an indicator variable); residing in a non-SMSA (rural area) reduces wages by 18.6%; and residing in the South reduces wages by 9.9%.

◀

► Example 2: Fixed-effects model

To fit the same model with the fixed-effects estimator, we specify the `fe` option.

```
. xtreg ln_w grade age c.age#c.age ttl_exp c.ttl_exp#c.ttl_exp tenure
> c.tenure#c.tenure 2.race not_smsa south, fe
note: grade omitted because of collinearity
note: 2.race omitted because of collinearity

Fixed-effects (within) regression              Number of obs   =   28091
Group variable: idcode                       Number of groups  =    4697

R-sq:  within = 0.1727                      Obs per group: min =     1
        between = 0.3505                      avg           =    6.0
        overall = 0.2625                      max           =    15

                                           F(8,23386)      =   610.12
corr(u_i, Xb) = 0.1936                      Prob > F         =    0.0000
```

ln_wage	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
grade	0 (omitted)					
age	.0359987	.0033864	10.63	0.000	.0293611	.0426362
c.age#c.age	-.000723	.0000533	-13.58	0.000	-.0008274	-.0006186
ttl_exp	.0334668	.0029653	11.29	0.000	.0276545	.039279
c.ttl_exp#c.ttl_exp	.0002163	.0001277	1.69	0.090	-.0000341	.0004666
tenure	.0357539	.0018487	19.34	0.000	.0321303	.0393775
c.tenure#c.tenure	-.0019701	.000125	-15.76	0.000	-.0022151	-.0017251
race						
black	0 (omitted)					
not_smsa	-.0890108	.0095316	-9.34	0.000	-.1076933	-.0703282
south	-.0606309	.0109319	-5.55	0.000	-.0820582	-.0392036
_cons	1.03732	.0485546	21.36	0.000	.9421496	1.13249
sigma_u	.35562203					
sigma_e	.29068923					
rho	.59946283	(fraction of variance due to u_i)				

```
F test that all u_i=0:      F(4696, 23386) =      6.65      Prob > F = 0.0000
```

The observation summary at the top is the same as for the between-effects model, although this time it is the “Number of obs” that is relevant.

Our three R^2 s are not too different from those reported previously; the R^2 within is slightly higher (0.1727 versus 0.1591), and the R^2 between is a little lower (0.3505 versus 0.4900), as expected, because the between estimator maximizes R^2 between and the within estimator R^2 within. In terms of overall fit, these estimates are somewhat worse (0.2625 versus 0.3695).

`xtreg, fe` can estimate σ_ν and σ_ϵ , although how you interpret these estimates depends on whether you are using `xtreg` to fit a fixed-effects model or random-effects model. To clarify this fine point, in the fixed-effects model, ν_i are formally fixed—they have no distribution. If you subscribe to this view, think of the reported $\hat{\sigma}_\nu$ as merely an arithmetic way to describe the range of the estimated but fixed ν_i . If, however, you are using the fixed-effects estimator of the random-effects model, 0.355622 is an estimate of σ_ν or would be if there were no omitted variables.

Here both `grade` and `2.race` were omitted from the model because they do not vary over time. Because `grade` and `2.race` are time invariant, our estimate u_i is an estimate of ν_i plus the effects of `grade` and `2.race`, so our estimate of the standard deviation is based on the variation in ν_i , `grade`, and `2.race`. On the other hand, had `2.race` and `grade` been omitted merely because they were collinear with the other regressors in our model, u_i would be an estimate of ν_i , and 0.355622 would be an estimate of σ_ν . (`xtsum` and `xttab` allow you to determine whether a variable is time invariant; see [XT] `xtsum` and [XT] `xttab`.)

Regardless of the status of u_i , our estimate of the standard deviation of ϵ_{it} is valid (and, in fact, is the estimate that would be used by the random-effects estimator to produce its results).

Our estimate of the correlation of u_i with $\mathbf{x}_{it}$ suffers from the problem of what u_i measures. We find correlation but cannot say whether this is correlation of ν_i with $\mathbf{x}_{it}$ or merely correlation of `grade` and `2.race` with $\mathbf{x}_{it}$. In any case, the fixed-effects estimator is robust to such a correlation, and the other estimates it produces are unbiased.

So, although this estimator produces no estimates of the effects of `grade` and `2.race`, it does predict that age has a positive effect on wages up to age 24.9 years (compared with 26.9 years estimated by the between estimator); that total experience still increases wages at an increasing rate (which is still bothersome); that tenure increases wages up to 9.1 years (compared with 12.1); that living in a non-SMSA reduces wages by 8.9% (compared with a more drastic 18.6%); and that living in the South reduces wages by 6.1% (as compared with 9.9%).

◀

► Example 3: Fixed-effects models with robust standard errors

If we suspect that there is heteroskedasticity or within-panel serial correlation in the idiosyncratic error term ϵ_{it} , we could specify the `vce(robust)` option:

```
. xtreg ln_w grade age c.age#c.age ttl_exp c.ttl_exp#c.ttl_exp tenure
> c.tenure#c.tenure 2.race not_smsa south, fe vce(robust)
note: grade omitted because of collinearity
note: 2.race omitted because of collinearity
Fixed-effects (within) regression      Number of obs      =      28091
Group variable: idcode                Number of groups    =      4697
R-sq:  within = 0.1727                 Obs per group: min =        1
      between = 0.3505                 avg               =       6.0
      overall  = 0.2625                 max               =      15
                                         F(8,4696)          =      273.86
corr(u_i, Xb) = 0.1936                 Prob > F            =      0.0000
                                         (Std. Err. adjusted for 4697 clusters in idcode)
```

ln_wage	Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]	
grade	0	(omitted)				
age	.0359987	.0052407	6.87	0.000	.0257243	.046273
c.age#c.age	-.000723	.0000845	-8.56	0.000	-.0008887	-.0005573
ttl_exp	.0334668	.004069	8.22	0.000	.0254896	.0414439
c.ttl_exp#c.ttl_exp	.0002163	.0001763	1.23	0.220	-.0001294	.0005619
tenure	.0357539	.0024683	14.49	0.000	.0309148	.040593
c.tenure#c.tenure	-.0019701	.0001696	-11.62	0.000	-.0023026	-.0016376
race						
black	0	(omitted)				
not_smsa	-.0890108	.0137629	-6.47	0.000	-.1159926	-.062029
south	-.0606309	.0163366	-3.71	0.000	-.0926583	-.0286035
_cons	1.03732	.0739644	14.02	0.000	.8923149	1.182325
sigma_u	.35562203					
sigma_e	.29068923					
rho	.59946283	(fraction of variance due to u_i)				

Although the estimated coefficients are the same with and without the `vce(robust)` option, the robust estimator produced larger standard errors and a p -value for `c.ttl_exp#c.ttl_exp` above the conventional 10%. The F test of $\nu_i = 0$ is suppressed because it is too difficult to compute the robust form of the statistic when there are more than a few panels.

❑ Technical note

The robust standard errors reported above are identical to those obtained by clustering on the panel variable `idcode`. Clustering on the panel variable produces an estimator of the VCE that is robust to cross-sectional heteroskedasticity and within-panel (serial) correlation that is asymptotically equivalent to that proposed by [Arellano \(1987\)](#). Although the example above applies the fixed-effects estimator, the robust and cluster-robust VCE estimators are also available for the random-effects estimator. [Wooldridge \(2013\)](#) and [Arellano \(2003\)](#) discuss these robust and cluster-robust VCE estimators for the fixed-effects and random-effects estimators. More details are available in [Methods and formulas](#).

➤ **Example 4: Random-effects model**

Refitting our log-wage model with the random-effects estimator, we obtain

```
. xtreg ln_w grade age c.age#c.age ttl_exp c.ttl_exp#c.ttl_exp tenure
> c.tenure#c.tenure 2.race not_smsa south, re theta

Random-effects GLS regression              Number of obs      =      28091
Group variable: idcode                    Number of groups     =       4697
R-sq:  within = 0.1715                    Obs per group: min =         1
      between = 0.4784                      avg =         6.0
      overall  = 0.3708                      max =        15

corr(u_i, X) = 0 (assumed)                Wald chi2(10)       =    9244.74
                                           Prob > chi2         =     0.0000
```

	min	5%	theta	95%	max
	0.2520	0.2520	0.5499	0.7016	0.7206

ln_wage	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
grade	.0646499	.0017812	36.30	0.000	.0611589	.0681409
age	.0368059	.0031195	11.80	0.000	.0306918	.0429201
c.age#c.age	-.0007133	.00005	-14.27	0.000	-.0008113	-.0006153
ttl_exp	.0290208	.002422	11.98	0.000	.0242739	.0337678
c.ttl_exp#						
c.ttl_exp	.0003049	.0001162	2.62	0.009	.000077	.0005327
tenure	.0392519	.0017554	22.36	0.000	.0358113	.0426925
c.tenure#						
c.tenure	-.0020035	.0001193	-16.80	0.000	-.0022373	-.0017697
race						
black	-.053053	.0099926	-5.31	0.000	-.0726381	-.0334679
not_smsa	-.1308252	.0071751	-18.23	0.000	-.1448881	-.1167622
south	-.0868922	.0073032	-11.90	0.000	-.1012062	-.0725781
_cons	.2387207	.049469	4.83	0.000	.1417633	.3356781
sigma_u	.25790526					
sigma_e	.29068923					
rho	.44045273	(fraction of variance due to u_i)				

According to the R^2 s, this estimator performs worse within than the within fixed-effects estimator and worse between than the between estimator, as it must, and slightly better overall.

We estimate that σ_ν is 0.2579 and σ_ϵ is 0.2907 and, by assertion, assume that the correlation of ν and $\mathbf{x}$ is zero.

All that is known about the random-effects estimator is its asymptotic properties, so rather than reporting an F statistic for overall significance, `xtreg, re` reports a χ^2 . Taken jointly, our coefficients are significant.

`xtreg, re` also reports a summary of the distribution of θ_i , an ingredient in the estimation of (4). θ is not a constant here because we observe women for unequal periods.

We estimate that schooling has a rate of return of 6.5% (compared with 6.1% between and no estimate within); that the increase of wages with age turns around at 25.8 years (compared with 26.9 between and 24.9 within); that total experience yet again increases wages increasingly; that the effect

of job tenure turns around at 9.8 years (compared with 12.1 between and 9.1 within); that being black reduces wages by 5.3% (compared with 5.6% between and no estimate within); that living in a non-SMSA reduces wages 13.1% (compared with 18.6% between and 8.9% within); and that living in the South reduces wages 8.7% (compared with 9.9% between and 6.1% within).

◀

► Example 5: Random-effects model fit using ML

We could also have fit this random-effects model with the maximum likelihood estimator:

```
. xtreg ln_w grade age c.age#c.age ttl_exp c.ttl_exp#c.ttl_exp tenure
> c.tenure#c.tenure 2.race not_smsa south, mle
```

Fitting constant-only model:

```
Iteration 0:  log likelihood = -13690.161
Iteration 1:  log likelihood = -12819.317
Iteration 2:  log likelihood = -12662.039
Iteration 3:  log likelihood = -12649.744
Iteration 4:  log likelihood = -12649.614
Iteration 5:  log likelihood = -12649.614
```

Fitting full model:

```
Iteration 0:  log likelihood = -8922.145
Iteration 1:  log likelihood = -8853.6409
Iteration 2:  log likelihood = -8853.4255
Iteration 3:  log likelihood = -8853.4254
```

Random-effects ML regression	Number of obs	=	28091
Group variable: idcode	Number of groups	=	4697
Random effects u_i ~ Gaussian	Obs per group: min	=	1
	avg	=	6.0
	max	=	15
	LR chi2(10)	=	7592.38
Log likelihood = -8853.4254	Prob > chi2	=	0.0000

ln_wage	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
grade	.0646093	.0017372	37.19	0.000	.0612044	.0680142
age	.0368531	.0031226	11.80	0.000	.030733	.0429732
c.age#c.age	-.0007132	.0000501	-14.24	0.000	-.0008113	-.000615
ttl_exp	.0288196	.0024143	11.94	0.000	.0240877	.0335515
c.ttl_exp# c.ttl_exp	.000309	.0001163	2.66	0.008	.0000811	.0005369
tenure	.0394371	.0017604	22.40	0.000	.0359868	.0428875
c.tenure# c.tenure	-.0020052	.0001195	-16.77	0.000	-.0022395	-.0017709
race						
black	-.0533394	.0097338	-5.48	0.000	-.0724172	-.0342615
not_smsa	-.1323433	.0071322	-18.56	0.000	-.1463221	-.1183644
south	-.0875599	.0072143	-12.14	0.000	-.1016998	-.0734201
_cons	.2390837	.0491902	4.86	0.000	.1426727	.3354947
/sigma_u	.2485556	.0035017			.2417863	.2555144
/sigma_e	.2918458	.001352			.289208	.2945076
rho	.4204033	.0074828			.4057959	.4351212

Likelihood-ratio test of sigma_u=0: chibar2(01)= 7339.84 Prob>=chibar2 = 0.000

The estimates are nearly the same as those produced by `xtreg, re`—the GLS estimator. For instance, `xtreg, re` estimated the coefficient on `grade` to be 0.0646499, `xtreg, mle` estimated 0.0646093, and the ratio is $0.0646499/0.0646093 = 1.001$ to three decimal places. Similarly, the standard errors are nearly equal: $0.0017811/0.0017372 = 1.025$. Below we compare all 11 coefficients:

Estimator	Coefficient ratio			SE ratio		
	mean	min.	max.	mean	min.	max.
xtreg, mle (ML)	1.	1.	1.	1.	1.	1.
xtreg, re (GLS)	.997	.987	1.007	1.006	.997	1.027



➤ Example 6: Population-averaged model

We could also have fit this model with the population-averaged estimator:

```
. xtreg ln_w grade age c.age#c.age ttl_exp c.ttl_exp#c.ttl_exp tenure
> c.tenure#c.tenure 2.race not_smsa south, pa

Iteration 1: tolerance = .0310561
Iteration 2: tolerance = .00074898
Iteration 3: tolerance = .0000147
Iteration 4: tolerance = 2.880e-07

GEE population-averaged model
Group variable:                idcode      Number of obs      =      28091
Link:                    identity      Number of groups      =      4697
Family:                  Gaussian      Obs per group: min     =          1
Correlation:             exchangeable      avg              =          6.0
                                      max              =          15
Scale parameter:                .1436709      Wald chi2(10)         =      9598.89
                                      Prob > chi2              =      0.0000
```

ln_wage	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
grade	.0645427	.0016829	38.35	0.000	.0612442	.0678412
age	.036932	.0031509	11.72	0.000	.0307564	.0431076
c.age#c.age	-.0007129	.0000506	-14.10	0.000	-.0008121	-.0006138
ttl_exp	.0284878	.0024169	11.79	0.000	.0237508	.0332248
c.ttl_exp#						
c.ttl_exp	.0003158	.0001172	2.69	0.007	.0000086	.0005456
tenure	.0397468	.0017779	22.36	0.000	.0362621	.0432315
c.tenure#						
c.tenure	-.002008	.0001209	-16.61	0.000	-.0022449	-.0017711
race						
black	-.0538314	.0094086	-5.72	0.000	-.072272	-.0353909
not_smsa	-.1347788	.0070543	-19.11	0.000	-.1486049	-.1209522
south	-.0885969	.0071132	-12.46	0.000	-.1025386	-.0746552
_cons	.2396286	.0491465	4.88	0.000	.1433034	.3359539

These results differ from those produced by `xtreg, re` and `xtreg, mle`. Coefficients are larger and standard errors smaller. `xtreg, pa` is simply another way to run the `xtgee` command. That is, we would have obtained the same output had we typed

```
. xtgee ln_w grade age c.age#c.age ttl_exp c.ttl_exp#c.ttl_exp
> tenure c.tenure#c.tenure 2.race not_smsa south
(output omitted because it is the same as above)
```

See [XT] [xtgee](#). In the language of [xtgee](#), the random-effects model corresponds to an exchangeable correlation structure and identity link, and [xtgee](#) also allows other correlation structures. Let's stay with the random-effects model, however. [xtgee](#) will also produce robust estimates of variance, and we refit this model that way by typing

```
. xtgee ln_w grade age c.age#c.age ttl_exp c.ttl_exp#c.ttl_exp
> tenure c.tenure#c.tenure 2.race not_smsa south, vce(robust)
(output omitted, coefficients the same, standard errors different)
```

In the [previous example](#), we presented a table comparing [xtreg, re](#) with [xtreg, mle](#). Below we add the results from the estimates shown and the ones we did with [xtgee, vce\(robust\)](#):

Estimator		Coefficient ratio			SE ratio		
		mean	min.	max.	mean	min.	max.
xtreg, mle	(ML)	1.	1.	1.	1.	1.	1.
xtreg, re	(GLS)	.997	.987	1.007	1.006	.997	1.027
xtreg, pa	(PA)	1.060	.847	1.317	.853	.626	.986
xtgee, vce(robust)	(PA)	1.060	.847	1.317	1.306	.957	1.545

So, which are right? This is a real dataset, and we do not know. However, in [example 2](#) in [XT] [xtreg postestimation](#), we will present evidence that the assumptions underlying the [xtreg, re](#) and [xtreg, mle](#) results are not met.



Stored results

`xtreg`, `re` stores the following in `e()`:

Scalars

<code>e(N)</code>	number of observations
<code>e(N_g)</code>	number of groups
<code>e(df_m)</code>	model degrees of freedom
<code>e(g_min)</code>	smallest group size
<code>e(g_avg)</code>	average group size
<code>e(g_max)</code>	largest group size
<code>e(Tcon)</code>	1 if T is constant
<code>e(sigma)</code>	ancillary parameter (γ , $\lnormal$)
<code>e(sigma_u)</code>	panel-level standard deviation
<code>e(sigma_e)</code>	standard deviation of ϵ_{it}
<code>e(r2_w)</code>	R -squared for within model
<code>e(r2_o)</code>	R -squared for overall model
<code>e(r2_b)</code>	R -squared for between model
<code>e(N_clust)</code>	number of clusters
<code>e(chi2)</code>	χ^2
<code>e(p)</code>	significance
<code>e(rho)</code>	ρ
<code>e(thta_min)</code>	minimum θ
<code>e(thta_5)</code>	θ , 5th percentile
<code>e(thta_50)</code>	θ , 50th percentile
<code>e(thta_95)</code>	θ , 95th percentile
<code>e(thta_max)</code>	maximum θ
<code>e(rmse)</code>	root mean squared error of GLS regression
<code>e(Tbar)</code>	harmonic mean of group sizes
<code>e(rank)</code>	rank of <code>e(V)</code>

Macros

<code>e(cmd)</code>	<code>xtreg</code>
<code>e(cmdline)</code>	command as typed
<code>e(depvar)</code>	name of dependent variable
<code>e(ivar)</code>	variable denoting groups
<code>e(model)</code>	<code>re</code>
<code>e(clustvar)</code>	name of cluster variable
<code>e(chi2type)</code>	Wald; type of model χ^2 test
<code>e(vce)</code>	<code>vcetype</code> specified in <code>vce()</code>
<code>e(vcetype)</code>	title used to label Std. Err.
<code>e(sa)</code>	Swamy–Arora estimator of the variance components (<code>sa</code> only)
<code>e(properties)</code>	<code>b V</code>
<code>e(predict)</code>	program used to implement <code>predict</code>
<code>e(marginsnotok)</code>	predictions disallowed by <code>margins</code>
<code>e(asbalanced)</code>	factor variables <code>fvset</code> as <code>asbalanced</code>
<code>e(asobserved)</code>	factor variables <code>fvset</code> as <code>asobserved</code>

Matrices

<code>e(b)</code>	coefficient vector
<code>e(bf)</code>	coefficient vector for fixed-effects model
<code>e(theta)</code>	θ
<code>e(V)</code>	variance–covariance matrix of the estimators
<code>e(VCEf)</code>	VCE for fixed-effects model

Functions

<code>e(sample)</code>	marks estimation sample
------------------------	-------------------------

xtreg, be stores the following in `e()`:

Scalars

<code>e(N)</code>	number of observations
<code>e(N_g)</code>	number of groups
<code>e(typ)</code>	WLS, if <code>wls</code> specified
<code>e(mss)</code>	model sum of squares
<code>e(df_m)</code>	model degrees of freedom
<code>e(rss)</code>	residual sum of squares
<code>e(df_r)</code>	residual degrees of freedom
<code>e(ll)</code>	log likelihood
<code>e(ll_0)</code>	log likelihood, constant-only model
<code>e(g_min)</code>	smallest group size
<code>e(g_avg)</code>	average group size
<code>e(g_max)</code>	largest group size
<code>e(Tcon)</code>	1 if T is constant
<code>e(r2)</code>	R -squared
<code>e(r2_a)</code>	adjusted R -squared
<code>e(r2_w)</code>	R -squared for within model
<code>e(r2_o)</code>	R -squared for overall model
<code>e(r2_b)</code>	R -squared for between model
<code>e(F)</code>	F statistic
<code>e(rmse)</code>	root mean squared error
<code>e(Tbar)</code>	harmonic mean of group sizes
<code>e(rank)</code>	rank of <code>e(V)</code>

Macros

<code>e(cmd)</code>	xtreg
<code>e(cmdline)</code>	command as typed
<code>e(depvar)</code>	name of dependent variable
<code>e(ivar)</code>	variable denoting groups
<code>e(model)</code>	be
<code>e(title)</code>	title in estimation output
<code>e(vce)</code>	<i>vcetype</i> specified in <code>vce()</code>
<code>e(vcetype)</code>	title used to label Std. Err.
<code>e(properties)</code>	b V
<code>e(predict)</code>	program used to implement predict
<code>e(marginsok)</code>	predictions allowed by <code>margins</code>
<code>e(marginsnotok)</code>	predictions disallowed by <code>margins</code>
<code>e(asbalanced)</code>	factor variables <code>fvset</code> as <code>asbalanced</code>
<code>e(asobserved)</code>	factor variables <code>fvset</code> as <code>asobserved</code>

Matrices

<code>e(b)</code>	coefficient vector
<code>e(V)</code>	variance–covariance matrix of the estimators

Functions

<code>e(sample)</code>	marks estimation sample
------------------------	-------------------------

`xtreg`, `fe` stores the following in `e()`:

Scalars

<code>e(N)</code>	number of observations
<code>e(N_g)</code>	number of groups
<code>e(mss)</code>	model sum of squares
<code>e(df_m)</code>	model degrees of freedom
<code>e(rss)</code>	residual sum of squares
<code>e(df_r)</code>	residual degrees of freedom
<code>e(tss)</code>	total sum of squares
<code>e(g_min)</code>	smallest group size
<code>e(g_avg)</code>	average group size
<code>e(g_max)</code>	largest group size
<code>e(Tcon)</code>	1 if T is constant
<code>e(sigma)</code>	ancillary parameter (γ , $\lnormal$)
<code>e(corr)</code>	$\text{corr}(u_i, Xb)$
<code>e(sigma_u)</code>	panel-level standard deviation
<code>e(sigma_e)</code>	standard deviation of ϵ_{it}
<code>e(r2)</code>	R -squared
<code>e(r2_a)</code>	adjusted R -squared
<code>e(r2_w)</code>	R -squared for within model
<code>e(r2_o)</code>	R -squared for overall model
<code>e(r2_b)</code>	R -squared for between model
<code>e(ll)</code>	log likelihood
<code>e(ll_0)</code>	log likelihood, constant-only model
<code>e(N_clust)</code>	number of clusters
<code>e(rho)</code>	ρ
<code>e(F)</code>	F statistic
<code>e(F_f)</code>	F for $u_i=0$
<code>e(df_a)</code>	degrees of freedom for absorbed effect
<code>e(df_b)</code>	numerator degrees of freedom for F statistic
<code>e(rmse)</code>	root mean squared error
<code>e(Tbar)</code>	harmonic mean of group sizes
<code>e(rank)</code>	rank of <code>e(V)</code>

Macros

<code>e(cmd)</code>	<code>xtreg</code>
<code>e(cmdline)</code>	command as typed
<code>e(depvar)</code>	name of dependent variable
<code>e(ivar)</code>	variable denoting groups
<code>e(model)</code>	<code>fe</code>
<code>e(wtype)</code>	weight type
<code>e(wexp)</code>	weight expression
<code>e(clustvar)</code>	name of cluster variable
<code>e(vce)</code>	<i>vcetype</i> specified in <code>vce()</code>
<code>e(vcetype)</code>	title used to label Std. Err.
<code>e(properties)</code>	<code>b V</code>
<code>e(predict)</code>	program used to implement <code>predict</code>
<code>e(marginsnotok)</code>	predictions disallowed by <code>margins</code>
<code>e(asbalanced)</code>	factor variables <code>fvset</code> as <code>asbalanced</code>
<code>e(asobserved)</code>	factor variables <code>fvset</code> as <code>asobserved</code>

Matrices

<code>e(b)</code>	coefficient vector
<code>e(Cns)</code>	constraints matrix
<code>e(V)</code>	variance–covariance matrix of the estimators

Functions

<code>e(sample)</code>	marks estimation sample
------------------------	-------------------------

`xtreg`, `mle` stores the following in `e()`:

Scalars

<code>e(N)</code>	number of observations
<code>e(N_g)</code>	number of groups
<code>e(df_m)</code>	model degrees of freedom
<code>e(g_min)</code>	smallest group size
<code>e(g_avg)</code>	average group size
<code>e(g_max)</code>	largest group size
<code>e(sigma_u)</code>	panel-level standard deviation
<code>e(sigma_e)</code>	standard deviation of ϵ_{it}
<code>e(ll)</code>	log likelihood
<code>e(ll_0)</code>	log likelihood, constant-only model
<code>e(ll_c)</code>	log likelihood, comparison model
<code>e(chi2)</code>	χ^2
<code>e(chi2_c)</code>	χ^2 for comparison test
<code>e(rho)</code>	ρ
<code>e(rank)</code>	rank of <code>e(V)</code>

Macros

<code>e(cmd)</code>	<code>xtreg</code>
<code>e(cmdline)</code>	command as typed
<code>e(depvar)</code>	name of dependent variable
<code>e(ivar)</code>	variable denoting groups
<code>e(model)</code>	<code>ml</code>
<code>e(wtype)</code>	weight type
<code>e(wexp)</code>	weight expression
<code>e(title)</code>	title in estimation output
<code>e(vce)</code>	<i>vcetype</i> specified in <code>vce()</code>
<code>e(vcetype)</code>	title used to label Std. Err.
<code>e(chi2type)</code>	Wald or LR; type of model χ^2 test
<code>e(chi2_ct)</code>	Wald or LR; type of model χ^2 test corresponding to <code>e(chi2_c)</code>
<code>e(distrib)</code>	Gaussian; the distribution of the RE
<code>e(properties)</code>	<code>b V</code>
<code>e(predict)</code>	program used to implement <code>predict</code>
<code>e(marginsnotok)</code>	predictions disallowed by <code>margins</code>
<code>e(asbalanced)</code>	factor variables <code>fvset</code> as <code>asbalanced</code>
<code>e(asobserved)</code>	factor variables <code>fvset</code> as <code>asobserved</code>

Matrices

<code>e(b)</code>	coefficient vector
<code>e(V)</code>	variance–covariance matrix of the estimators

Functions

<code>e(sample)</code>	marks estimation sample
------------------------	-------------------------

`xtreg`, `pa` stores the following in `e()`:

Scalars

<code>e(N)</code>	number of observations
<code>e(N_g)</code>	number of groups
<code>e(df_m)</code>	model degrees of freedom
<code>e(chi2)</code>	χ^2
<code>e(p)</code>	significance
<code>e(df_pear)</code>	degrees of freedom for Pearson χ^2
<code>e(chi2_dev)</code>	χ^2 test of deviance
<code>e(chi2_dis)</code>	χ^2 test of deviance dispersion
<code>e(deviance)</code>	deviance
<code>e(dispers)</code>	deviance dispersion
<code>e(phi)</code>	scale parameter
<code>e(g_min)</code>	smallest group size
<code>e(g_avg)</code>	average group size
<code>e(g_max)</code>	largest group size
<code>e(rank)</code>	rank of <code>e(V)</code>
<code>e(tol)</code>	target tolerance
<code>e(dif)</code>	achieved tolerance
<code>e(rc)</code>	return code

Macros

<code>e(cmd)</code>	<code>xtgee</code>
<code>e(cmd2)</code>	<code>xtreg</code>
<code>e(cmdline)</code>	command as typed
<code>e(depvar)</code>	name of dependent variable
<code>e(ivar)</code>	variable denoting groups
<code>e(tvar)</code>	variable denoting time within groups
<code>e(model)</code>	<code>pa</code>
<code>e(family)</code>	Gaussian
<code>e(link)</code>	identity; link function
<code>e(corr)</code>	correlation structure
<code>e(scale)</code>	<code>x2</code> , <code>dev</code> , <code>phi</code> , or <code>#</code> ; scale parameter
<code>e(wtype)</code>	weight type
<code>e(wexp)</code>	weight expression
<code>e(offset)</code>	linear offset variable
<code>e(chi2type)</code>	Wald; type of model χ^2 test
<code>e(vce)</code>	<code>vce</code> type specified in <code>vce()</code>
<code>e(vcetype)</code>	title used to label Std. Err.
<code>e(rgf)</code>	<code>rgf</code> , if <code>rgf</code> specified
<code>e(nmp)</code>	<code>nmp</code> , if specified
<code>e(properties)</code>	<code>b V</code>
<code>e(predict)</code>	program used to implement <code>predict</code>
<code>e(marginsnotok)</code>	predictions disallowed by <code>margins</code>
<code>e(asbalanced)</code>	factor variables <code>fvset</code> as <code>asbalanced</code>
<code>e(asobserved)</code>	factor variables <code>fvset</code> as <code>asobserved</code>

Matrices

<code>e(b)</code>	coefficient vector
<code>e(R)</code>	estimated working correlation matrix
<code>e(V)</code>	variance-covariance matrix of the estimators
<code>e(V_modelbased)</code>	model-based variance

Functions

<code>e(sample)</code>	marks estimation sample
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Methods and formulas

The model to be fit is

$$y_{it} = \alpha + \mathbf{x}_{it}\beta + \nu_i + \epsilon_{it}$$

for $i = 1, \dots, n$ and, for each i , $t = 1, \dots, T$, of which T_i periods are actually observed.

Methods and formulas are presented under the following headings:

xtreg, fe
 xtreg, be
 xtreg, re
 xtreg, mle
 xtreg, pa

xtreg, fe

xtreg, fe produces estimates by running OLS on

$$(y_{it} - \bar{y}_i + \bar{\bar{y}}) = \alpha + (\mathbf{x}_{it} - \bar{\mathbf{x}}_i + \bar{\bar{\mathbf{x}}})\beta + (\epsilon_{it} - \bar{\epsilon}_i + \bar{\bar{\nu}}) + \bar{\bar{\epsilon}}$$

where $\bar{y}_i = \sum_{t=1}^{T_i} y_{it}/T_i$, and similarly, $\bar{\bar{y}} = \sum_i \sum_t y_{it}/(nT_i)$. The conventional covariance matrix of the estimators is adjusted for the extra $n - 1$ estimated means, so results are the same as using OLS on (1) to estimate ν_i directly. Specifying `vce(robust)` or `vce(cluster clustvar)` causes the Huber/White/sandwich VCE estimator to be calculated for the coefficients estimated in this regression. See [P] [_robust](#), particularly [Introduction](#) and [Methods and formulas](#). Wooldridge (2013) and Arellano (2003) discuss this application of the Huber/White/sandwich VCE estimator. As discussed by Wooldridge (2013), Stock and Watson (2008), and Arellano (2003), specifying `vce(robust)` is equivalent to specifying `vce(cluster panelvar)`, where *panelvar* is the variable that identifies the panels.

Clustering on the panel variable produces a consistent VCE estimator when the disturbances are not identically distributed over the panels or there is serial correlation in ϵ_{it} .

The cluster-robust VCE estimator requires that there are many clusters and the disturbances are uncorrelated across the clusters. The panel variable must be nested within the cluster variable because of the within-panel correlation induced by the within transform.

From the estimates $\hat{\alpha}$ and $\hat{\beta}$, estimates u_i of ν_i are obtained as $u_i = \bar{y}_i - \hat{\alpha} - \bar{\mathbf{x}}_i\hat{\beta}$. Reported from the calculated u_i are its standard deviation and its correlation with $\bar{\mathbf{x}}_i\hat{\beta}$. Reported as the standard deviation of e_{it} is the regression's estimated root mean squared error, s , which is adjusted (as previously stated) for the $n - 1$ estimated means.

Reported as R^2 within is the R^2 from the mean-deviated regression.

Reported as R^2 between is $\text{corr}(\bar{\mathbf{x}}_i\hat{\beta}, \bar{y}_i)^2$.

Reported as R^2 overall is $\text{corr}(\mathbf{x}_{it}\hat{\beta}, y_{it})^2$.

xtreg, be

xtreg, be fits the following model:

$$\bar{y}_i = \alpha + \bar{\mathbf{x}}_i\beta + \nu_i + \bar{\epsilon}_i$$

Estimation is via OLS unless T_i is not constant and the `wls` option is specified. Otherwise, the estimation is performed via WLS. The estimates and conventional VCE are obtained from `regress` for both cases, but for WLS, `[aweight=T_i]` is specified.

Reported as R^2 between is the R^2 from the fitted regression.

Reported as R^2 within is $\text{corr}\{(\mathbf{x}_{it} - \bar{\mathbf{x}}_i)\hat{\beta}, y_{it} - \bar{y}_i\}^2$.

Reported as R^2 overall is $\text{corr}(\mathbf{x}_{it}\hat{\beta}, y_{it})^2$.

xtreg, re

The key to the random-effects estimator is the GLS transform. Given estimates of the idiosyncratic component, $\hat{\sigma}_e^2$, and the individual component, $\hat{\sigma}_u^2$, the GLS transform of a variable z for the random-effects model is

$$z_{it}^* = z_{it} - \hat{\theta}_i \bar{z}_i$$

where $\bar{z}_i = \frac{1}{T_i} \sum_t^{T_i} z_{it}$ and

$$\hat{\theta}_i = 1 - \sqrt{\frac{\hat{\sigma}_e^2}{T_i \hat{\sigma}_u^2 + \hat{\sigma}_e^2}}$$

Given an estimate of $\hat{\theta}_i$, one transforms the dependent and independent variables, and then the coefficient estimates and the conventional variance–covariance matrix come from an OLS regression of y_{it}^* on $\mathbf{x}_{it}^*$ and the transformed constant $1 - \hat{\theta}_i$. Specifying `vce(robust)` or `vce(cluster clustvar)` causes the Huber/White/sandwich VCE estimator to be calculated for the coefficients estimated in this regression. See [P] [_robust](#); in particular, see [Introduction](#) and [Methods and formulas](#). [Wooldridge \(2013\)](#) and [Arellano \(2003\)](#) discuss this application of the Huber/White/sandwich VCE estimator. As discussed by [Wooldridge \(2013\)](#), [Stock and Watson \(2008\)](#), and [Arellano \(2003\)](#), specifying `vce(robust)` is equivalent to specifying `vce(cluster panelvar)`, where *panelvar* is the variable that identifies the panels.

Clustering on the panel variable produces a consistent VCE estimator when the disturbances are not identically distributed over the panels or there is serial correlation in ϵ_{it} .

The cluster–robust VCE estimator requires that there are many clusters and the disturbances are uncorrelated across the clusters. The panel variable must be nested within the cluster variable because of the within-panel correlation that is generally induced by the random-effects transform when there is heteroskedasticity or within-panel serial correlation in the idiosyncratic errors.

Stata has two implementations of the Swamy–Arora method for estimating the variance components. They produce the same results in balanced panels and share the same estimator of σ_e^2 . However, the two methods differ in their estimator of σ_u^2 in unbalanced panels. We call the first $\hat{\sigma}_{u\overline{T}}^2$ and the second $\hat{\sigma}_{u\text{SA}}^2$. Both estimators are consistent; however, $\hat{\sigma}_{u\text{SA}}^2$ has a more elaborate adjustment for small samples than $\hat{\sigma}_{u\overline{T}}^2$. (See [Baltagi \[2013\]](#), [Baltagi and Chang \[1994\]](#), and [Swamy and Arora \[1972\]](#) for derivations of these methods.)

Both methods use the same function of within residuals to estimate the idiosyncratic error component σ_e . Specifically,

$$\hat{\sigma}_e^2 = \frac{\sum_i^n \sum_t^{T_i} e_{it}^2}{N - n - K + 1}$$

where

$$e_{it} = (y_{it} - \bar{y}_i + \bar{\bar{y}}) - \hat{\alpha}_w - (\mathbf{x}_{it} - \bar{\mathbf{x}}_i + \bar{\bar{\mathbf{x}}})\hat{\beta}_w$$

and $\hat{\alpha}_w$ and $\hat{\beta}_w$ are the within estimates of the coefficients and $N = \sum_i^n T_i$. After passing the within residuals through the within transform, only the idiosyncratic errors are left.

The default method for estimating σ_u^2 is

$$\hat{\sigma}_{u\bar{T}}^2 = \max \left\{ 0, \frac{SSR_b}{n - K} - \frac{\hat{\sigma}_e^2}{\bar{T}} \right\}$$

where

$$SSR_b = \sum_i^n \left(\bar{y}_i - \hat{\alpha}_b - \bar{x}_i \hat{\beta}_b \right)^2$$

$\hat{\alpha}_b$ and $\hat{\beta}_b$ are coefficient estimates from the between regression and $\bar{T}$ is the harmonic mean of T_i :

$$\bar{T} = \frac{n}{\sum_i^n \frac{1}{T_i}}$$

This estimator is consistent for σ_u^2 and is computationally less expensive than the second method. The sum of squared residuals from the between model estimate a function of both the idiosyncratic component and the individual component. Using our estimator of σ_e^2 , we can remove the idiosyncratic component, leaving only the desired individual component.

The second method is the Swamy–Arora method for unbalanced panels derived by [Baltagi and Chang \(1994\)](#), which has a more precise small-sample adjustment. Using this method,

$$\hat{\sigma}_{uSA}^2 = \max \left\{ 0, \frac{SSR_b - (n - K)\hat{\sigma}_e^2}{N - tr} \right\}$$

where

$$tr = \text{trace} \{ (\mathbf{X}'\mathbf{P}\mathbf{X})^{-1} \mathbf{X}'\mathbf{Z}\mathbf{Z}'\mathbf{X} \}$$

$$\mathbf{P} = \text{diag} \left\{ \left(\frac{1}{T_i} \right) \boldsymbol{\nu}_{T_i} \boldsymbol{\nu}_{T_i}' \right\}$$

$$\mathbf{Z} = \text{diag} [\boldsymbol{\nu}_{T_i}]$$

$\mathbf{X}$ is the $N \times K$ matrix of covariates, including the constant, and $\boldsymbol{\nu}_{T_i}$ is a $T_i \times 1$ vector of ones.

The estimated coefficients $(\hat{\alpha}_r, \hat{\beta}_r)$ and their covariance matrix $\mathbf{V}_r$ are reported together with the previously calculated quantities $\hat{\sigma}_e$ and $\hat{\sigma}_u$. The standard deviation of $\nu_i + e_{it}$ is calculated as $\sqrt{\hat{\sigma}_e^2 + \hat{\sigma}_u^2}$.

Reported as R^2 between is $\text{corr}(\bar{x}_i \hat{\beta}, \bar{y}_i)^2$.

Reported as R^2 within is $\text{corr}\{(\mathbf{x}_{it} - \bar{x}_i) \hat{\beta}, y_{it} - \bar{y}_i\}^2$.

Reported as R^2 overall is $\text{corr}(\mathbf{x}_{it} \hat{\beta}, y_{it})^2$.

xtreg, mle

The log likelihood for the i th unit is

$$l_i = -\frac{1}{2} \left(\frac{1}{\sigma_e^2} \left[\sum_{t=1}^{T_i} (y_{it} - \mathbf{x}_{it}\beta)^2 - \frac{\sigma_u^2}{T_i\sigma_u^2 + \sigma_e^2} \left\{ \sum_{t=1}^{T_i} (y_{it} - \mathbf{x}_{it}\beta) \right\}^2 \right] \right. \\ \left. + \ln \left(T_i \frac{\sigma_u^2}{\sigma_e^2} + 1 \right) + T_i \ln(2\pi\sigma_e^2) \right)$$

The `mle` and `re` options yield essentially the same results, except when total $N = \sum_i T_i$ is small (200 or less) and the data are unbalanced.

xtreg, pa

See [XT] [xtgee](#) for details on the methods and formulas used to calculate the population-averaged model using a generalized estimating equations approach.

Acknowledgments

We thank Richard Goldstein, who wrote the first draft of the routine that fits random-effects regressions, Badi Baltagi of the Department of Economics at Syracuse University, and Manuelita Ureta of the Department of Economics at Texas A&M University, who assisted us in working our way through the literature.

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Also see

- [XT] **xtreg postestimation** — Postestimation tools for xtreg
- [XT] **xtgee** — Fit population-averaged panel-data models by using GEE
- [XT] **xtgls** — Fit panel-data models by using GLS
- [XT] **xtivreg** — Instrumental variables and two-stage least squares for panel-data models
- [XT] **xtregar** — Fixed- and random-effects linear models with an AR(1) disturbance
- [XT] **xtset** — Declare data to be panel data
- [ME] **mixed** — Multilevel mixed-effects linear regression
- [MI] **estimation** — Estimation commands for use with mi estimate
- [R] **areg** — Linear regression with a large dummy-variable set
- [R] **regress** — Linear regression
- [TS] **forecast** — Econometric model forecasting
- [TS] **prais** — Prais–Winsten and Cochrane–Orcutt regression
- [U] **20 Estimation and postestimation commands**